

Nils Duran

AI & Machine Learning Engineer | MSc Data Science Student

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Experience

Sparsity Technologies , Data Engineering Intern	Barcelona, Catalonia Oct 2025 – Jan 2026 4 months
- Engineered automated data collection and ingestion pipelines for high-performance graph database solutions. - Executed complex graph queries and data transformation workflows for competitor intelligence and marketing analytics.	
Werfen , Software Tester Intern — Beacon Project	Barcelona, Catalonia June 2026 – present 4 months
- Engineered full-stack automated testing suites in Java for in-vitro diagnostic (IVD) laboratory software on the Beacon Project. - Implemented comprehensive API tests and FormUI end-to-end automation, verifying data consistency across clinical workflows. - Managed requirements traceability, test planning, and regulatory compliance validation using Jama Connect. - Built custom utility scripts and automation tooling that eliminated repetitive manual QA tasks and accelerated test cycles.	

Education

MSc Universitat Politècnica de Catalunya (UPC) , Data Science	Barcelona, Catalonia Sept 2026 – present
Distributed data architectures, statistical modeling, and scalable machine learning.	
BSc Universitat Politècnica de Catalunya (UPC) , Artificial Intelligence	Barcelona, Catalonia Sept 2022 – June 2026
Thesis on EEG cognitive decoding using Spatio-Temporal GNNs, Transformers, and LoRA.	
Exchange Karlsruhe Institute of Technology (KIT) , Computer Science	Karlsruhe, Germany Oct 2024 – Mar 2025
Advanced distributed algorithms and high-performance systems. German C1 certified.	

Projects

Sinergia — Multi-Agent LLMs (Telefónica Colab)	May 2025
- Designed multi-expert LangGraph agent architecture with LoRA fine-tuning, boosting MedQA accuracy by +3% absolute over Gemini Flash. - Evaluated diversity dissimilarity matrices and Elo rankings; developed interactive Streamlit app (Public Repo).	
HPC Vision Transformers on MareNostrum 5 (BSC)	June 2024
- Trained Vision Transformers on the 200 GB+ MAME dataset using the MareNostrum 5 supercomputer (top 10 public supercomputer worldwide). - Orchestrated distributed multi-GPU training with Slurm and PyTorch, achieving a 0.75 F1-score across hundreds of fine-grained categories.	
Vegan Tools — Production AI Web Platform	Sept 2026
- Architected and deployed full-stack platform featuring Ingredient Checker, Menu OCR, and Recipe Vega-nizer (Live at vegantools.org). - Engineered end-to-end multi-modal AI API integration, client state management, and custom cloud infrastructure (Public Repo).	

Skills

AI Frameworks & LLMs: PyTorch, LangChain, LangGraph, LoRA / PEFT, Hugging Face Transformers, Gymnasium (RL), Scikit-learn
Engineering & Deployment: Python, Slurm (MareNostrum 5 HPC), Docker, Git/GitHub, REST APIs, Streamlit, Linux, Bash
Data & Backend: PostgreSQL, SQL, Graph Databases (Sparsity), Apache Spark, Vector Search, Automated Testing (Java)
Languages: Catalan (Native), Spanish (Bilingual), English (C2 Proficient), German (C1 Advanced)